2024 HCI Preliminary Examination Paper 2 Suggested Solutions

Q1		
(a)	During the collision, there are <u>no external forces acting on the photon and electron or system</u>, hence linear momentum is conserved.	**B1**
(b)(i)	**1.** $\sum p_x = 7.30 \times 10^{-22}$ kg m s^{-1}	**B1**
	2. $\sum p_Y = 0$ kg m s^{-1}	**B1**
(b)(ii)	By principle of conservation of linear momentum, $(\rightarrow)$ $7.3 \times 10^{-22} = (p_p)(\cos 60^o) + (p_e)(\cos 25^o)$ ...(1) $(\uparrow)$ $(p_e)(\sin 25^o) = (p_p)(\sin 60^o)$ ...(2) Solving (1) and (2) gives, $\frac{(p_p)(\sin 60°)}{(p_p)(\cos 60°)} = \tan 60° = \frac{(p_e)(\sin 25°)}{(7.3 \times 10^{-22}) - (p_e)(\cos 25°)}$ $\Rightarrow p_e = 6.35 \times 10^{-22}$ kg m s^{-1} M1, M1 – 2 equations showing application of COLM A1 – final answer after solving two equations.	**M1** **M1** **A1**